

Last time: Introduced context-free grammars.

Def: A context-free grammar (CFG) is a quadruple $G = (N, \Sigma, P, S)$.

- N - set of *nonterminals*
- Σ - alphabet of *terminal* characters
- P - *productions*. Pairs in $N \times (N \cup \Sigma)^*$.
- $S \in N$: Start symbol.

Warmup: Two modifications to $\{a^n b^n \mid n \geq 0\}$

- $S \rightarrow aSb \mid aS \mid \epsilon$
- $S \rightarrow aSb \mid C$
 $C \rightarrow cC \mid \epsilon$
- $SS \rightarrow aSa \mid bSb \mid \epsilon$

What are these languages?

How would you make the last grammar into one for the language of palindromes?

Derivations

Derivation: $\alpha \xRightarrow[G]{1} \beta$ if

- $\alpha = \alpha_1 A \alpha_2$ for $\alpha_1, \alpha_2 \in (N \cup \Sigma)^*$, $A \in N$
- $A \rightarrow \gamma \in P$
- $\alpha_1 \gamma \alpha_2 = \beta$

Idea: Can get from α to β in one step.

$\alpha \xRightarrow[G]{n+1} \beta$ if there exists some $\gamma \in (N \cup \Sigma)^*$ such that $\alpha \xRightarrow[G]{n} \gamma$ and $\gamma \xRightarrow[G]{1} \beta$.

$\alpha \xRightarrow[G]{*} \beta$ if $\alpha \xRightarrow[G]{n} \beta$ for some $n \geq 1$.

$$L(G) = \{x \in \Sigma^* \mid S \xRightarrow[G]{*} x\}$$

$$G_4. S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$$

Def. Let $\alpha \in (N \cup \Sigma)^*$. If $S \xRightarrow[G_3]{*} \alpha$, then α is a *sentential form* for G_3 .

Which are in $L(G_4)$?

- $aabaa$
- $aabbba$
- $ababab$

- $aabb$

What are the derivations for the ones that work?

Conjectures for what this language is?

Proving Correctness

Notation: Let $d(\alpha) = \#a(\alpha) - \#b(\alpha)$

Claim: Let $S \xrightarrow[G_4]{*} \alpha$. Then $d(\alpha) = 0$.

For intuition: Track $d(\alpha)$ through a derivation.

Idea: Every production keeps $d(\alpha)$ the same.

Proof structural induction.

Base Case: $S \xrightarrow[G_3]{0} \alpha$. i.e. $\alpha = S$.

Inductive Hypothesis: That for all sentential forms α derivable in n steps, $d(\alpha) = 0$.

Inductive Case: Assume $S \xrightarrow[G_4]{n+1} \alpha$.

By definition of $\xrightarrow[G_4]{n+1}$, $S \xrightarrow[G_4]{n} \beta S \gamma \xrightarrow[G_4]{1} \alpha$. (Point out in example.)

Inductive hypothesis: $d(\beta S \gamma) = 0$

Cases: One for each choice of final production.

- $S \rightarrow \epsilon$: $d(\beta S \gamma) = d(\beta \gamma) = d(\alpha)$
- $S \rightarrow aSb$: $d(\beta S \gamma) = d(\beta a S b \gamma) + 1 - 1 = d(\alpha)$
- $S \rightarrow bSa$: $d(\beta S \gamma) = d(\beta b S a \gamma) + 1 - 1 = d(\alpha)$
- $S \rightarrow SS$: $d(\beta S \gamma) = d(\beta SS \gamma) = d(\alpha)$

Less clear: Can we generate any string $x \in \{a, b\}^*$ with $d(x) = 0$?

Claim: If $x \in \{a, b\}^*$ and $d(x) = 0$, then $S \xrightarrow[G_4]{*} x$.

Proof: Strong induction on $|x|$.

Base Case: $|x| = 0$. $x = \epsilon$. $S \rightarrow \epsilon$.

Inductive Hypothesis: Assume for all $x \in \{a, b\}^*$ of length $\leq n$, if $d(x) = 0$, then $S \xrightarrow[G_4]{*} x$.

Inductive Case: Assume $|x| = n + 1$.

Trick: Plot $d(x_1 \dots x_i)$ for all i .

Example: $aabbbba$.

Case 1: Assume plot crosses 0

Then $x = yz$, with $d(y) = 0$ and $d(z) = 0$ and $|y|, |z| \geq 1$.

By IH: $S \xrightarrow[G_4]{*} y$, $S \xrightarrow[G_4]{*} z$, so

$$S \xrightarrow[G_4]{1} SS \xrightarrow[G_4]{*} yS \xrightarrow[G_4]{*} yz.$$

Case 2: Otherwise: Draw plots that don't cross 0.

$\Rightarrow$ What can we say about first/last character?

$\Rightarrow x = ayb$ or $x = bya$. What is $d(y)$?

$$S \xrightarrow[G_4]{1} aSb \xrightarrow[G_4]{*} ayb$$